Title: An open-source, provenance-aware framework for reconstructing missing CT dose indices from DICOM metadata

Running title: Provenance-aware CT dose reconstruction

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Conflicts of interest. The author is the representative of LISIT Co., Ltd. (Tokyo, Japan) and Chief Executive Officer of TexelCraft OÜ, and this work was conducted under the supervision of LISIT Co., Ltd. The software is released under the MIT licence and the author has a commercial interest in downstream products that may use it. No patient or customer data were used.

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Ethics approval and consent. Not applicable. This study used only publicly available, de-identified imaging data (TCIA collections released under Creative Commons licences) and third-party sample files. No institutional review board approval and no informed consent were required.

Data and code availability. The software is available at https://github.com/Institute-of-One/ctdose-core under the MIT License. The version used in this study (version 0.1.1) is archived at Zenodo (https://doi.org/10.5281/zenodo.21636719).

Supplementary Material. One supplementary file accompanies this Technical Note, containing the optional organ-dose layer, the detailed stratified analyses and Figures S1–S3.

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Author contributions. S.Y.: conceptualisation, methodology, software, validation, formal analysis, investigation, data curation, writing — original draft, writing — review and editing, visualisation.